

A Robot that can Dream, Recognise and Greet Visitors

A first-of-its-kind robot that can recognise a human face and dream—like a human—can now be yours! Interested?

We humans can speak, recognise feelings and emotions, and even dream, which makes us different from machines. But does it make us more intelligent than them? Can machines like robots also acquire this capability?

If yes, how can such intelligent robots be used? As artificial assistants in the workplace, as companions for the elderly and children, for welcoming guests in hotels; these intelligent robots can have multiple uses.

Very few present-day robots are intelligent and they are expensive, costing up to hundreds of thousands of rupees! Keeping this in mind, we are presenting here an intelligent robot that is based on open source tools and libraries. We have used an InMoov robot face mask. You can use the same or any other of your choice. The theme of the robot is shown in Figure 1, while the dreaming robot is shown in Figure 2.

Installation and working

Using Google's open source Deep Dream library, a camera takes a picture and then creates dreams based on it. Artificial intelligence (AI) and Python are used to make the robot recognise your face and talk to you.

Basically, this is the second version of the project that was published in November 2020 issue of *Electronics For You* magazine. So, for more information, please see the AI robot details there. This robot is more capable than the previous as it greets the person on recognising and informs us. It also streams the live video over VNC Viewer, as shown in Figure 3. What's more, it has an amazing capability to dream like a human being.

Bill of Materials

Components	Quantity	Description
Raspberry Pi 4	1	2Gb - 4Gb or Higher
Pi Camera With CSI Ribbon	1	5Mp CSI Ribbon Cable
SD Card, 16-32 GB	1	Class 10 or Higher
5V DC Power Adaptor	1	1.5A-2A DC Power Supply
3D Printed Robot Face	1	PLA/ABS 3D Printed Robot



Figure 1: The robot's theme

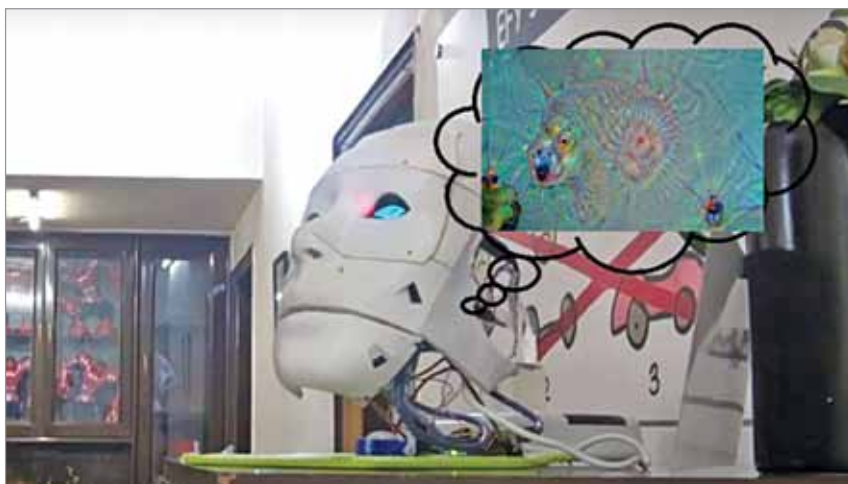


Figure 2: The dreaming robot